

INTELLIGENT TELEMEDICINE CONSULTATION SYSTEM

A Secure, Intelligent & Database-Driven Healthcare Platform

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Course

Database Systems Engineering and
Distributed Backend Development

ABSTRACT

The Intelligent Telemedicine Consultation System is a centralized healthcare platform that connects Patients, Doctors and Administrators on a single, secure environment for consultations, records and clinical decision support.



Patient
Books appointments, consults doctors, views records



Doctor
Manages consultations, prescriptions, patient history



Administrator
Oversees users, data integrity and system operations

CORE DATA MANAGED

Patient & Doctor Profiles

Appointments

Consultations

Prescriptions

Medical Reports

Health History

Notifications

Referrals

Backend & Data Layer: Node.js and Express.js provide the API layer; MongoDB with Mongoose deliver schema-based modeling, validation, relationships, querying, indexing and aggregation.

 AI Symptom Analysis

 Multilingual Translation

 Weather & Air Quality Info

 WHO Disease Alerts

 Govt. Health Advisories

Intelligent Telemedicine Consultation System

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PROBLEM STATEMENT

- Healthcare information is often distributed across different processes and touchpoints.
- Patient, doctor and appointment information need to remain connected and consistent.
- Medical records, prescriptions and consultation history require centralized management.
- Different healthcare users require controlled, role-based access to information.
- Growing healthcare data requires efficient storage, retrieval and analysis.
- Healthcare systems benefit from intelligent services that provide timely guidance.

CORE PROBLEM

How can a secure and centralized healthcare system efficiently manage interconnected patient and clinical information while supporting intelligent healthcare services?



EXISTING VS. PROPOSED SYSTEM

✗ EXISTING SYSTEM

- Healthcare information maintained across separate systems
- Limited integration between appointments, consultations and records
- Patient history difficult to access as a unified record
- Data analysis often requires separate manual processes
- Limited integration of intelligent healthcare services
- Role-based information management is often inconsistent

✓ PROPOSED SYSTEM

- Centralized telemedicine platform for Patients, Doctors & Admins
- Integrated management of appointments, consultations & records
- MongoDB + Mongoose for persistent, schema-based data storage
- Node.js + Express.js REST API backend with validation
- CRUD operations, filtering, indexing and aggregation
- Role-based access, authentication and AI-driven services

PROPOSED SYSTEM ARCHITECTURE



KEY DATABASE ENTITIES



TOOLS & TECHNOLOGIES USED



FRONTEND

- React.js
- JavaScript
- HTML5
- CSS3



BACKEND

- Node.js
- Express.js
- REST APIs



DATABASE

- MongoDB
- Mongoose



SECURITY

- JWT Authentication
- Password Hashing
- Role-Based Access



AI & EXTERNAL SERVICES

- Gemini AI
- OpenWeather API



DEVELOPMENT & TESTING

- Visual Studio Code
- Git / GitHub
- Postman